

Program Analysis 2025/26

Exercises – May 21, 2026

[Ex. 1] Let us consider the following axiom for atomic boolean guards $b?$, where the parentheses $\langle \cdot \rangle \in \{ \{ \cdot \}, [\cdot], (\cdot), \langle \cdot \rangle \}$ are used to denote a generic program logic triple of either Hoare logic, IL, NC or SIL.

$$\frac{}{\langle P \rangle b? \langle P \vee b \rangle}$$

Explain for which logics (HL, IL, NC, SIL) the axiom is sound (for any precondition P and boolean guard b):

- if the axiom is sound, prove its validity;
- if it is not sound, first provide a concrete counterexample and then say under which conditions on P and b the axiom would become sound.

[Ex. 2] Let us consider the usual concrete domain $(\wp(\mathbb{Z}), \subseteq)$ of sets of integers ordered by inclusion and the abstract domain $A = \{ \perp, = 0, \leq 0, > 0, > 10, \top \}$ with the obvious concretization function.

1. Define the GC maps $\gamma : A \rightarrow \wp(\mathbb{Z})$ and $\alpha : \wp(\mathbb{Z}) \rightarrow A$.
2. Draw the Hasse diagram of A .
3. Define the bca $\cdot \times^A \cdot$ for the product of integers $\cdot \times \cdot$.
4. Show that $\cdot \times^A \cdot$ is not complete by exhibiting a counterexample.

[Ex. 3] Consider the following expression e :

```
let apply = fn f => f 2
in let inc = fn x => x + 4
    in let dbl = fn y => y * 2
        in (apply inc) - (apply dbl)
```

1. Add labels to all subexpressions and function abstractions.
2. Guess a possible analysis result for Pure 0-CFA and verify it using the corresponding acceptability relation $\models$. In particular, comment on the abstract values associated with the two calls to `apply`.
3. Consider the following additional abstract flow:

$$\text{fn } w \Rightarrow w \in \hat{C}(\text{Label}(\mathbf{f}))$$

Does the acceptability relation $\models$ still hold? What does this say about the difference between soundness and precision?

4. Discuss how the analysis changes using the syntax-directed version of the acceptability relation $\models_s$.
5. Extract the corresponding system of constraints.
6. Explain how the result changes under 1-CFA. Which spurious flows disappear, and why?